# Manual Milling Machine

## Standard Operating Procedure

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## 1. Introduction

This document is to be used by properly trained and experienced students and staff as a guideline for the safe operation of this machine and is written in accordance with OSHA and PERRP standards.

## 2. Scope and Application

This manual vertical milling machine (knee mill) is to be used only for material removal and measurement purposes in service to the Engineering Student Projects teams housed in this facility, as well as for support of said facility. Students and Staff who wish to use this equipment must have first received training from the Shop Supervisor and have read and signed both the Motorsports Disclosure Form and the Machine Shop Access Form. This is described in greater detail in SECTION 6.

## 3. Responsibilities

- The Shop Supervisor will update and ensure the availability of this document at the machine.
- The Shop Supervisor (and staff when instructed) will solely be responsible for performing/arranging maintenance, repair and machine modifications to ensure the machine is in safe operation condition.
- Student Operator is required to have read, signed and turned in the appropriate forms and have participated in task specific training instructed by the Shop Supervisor before operating any equipment in the Facility. This is described in greater detail in Section 6.
- Student Operator is responsible for all cleaning and housekeeping. This machine must be kept in showroom condition as it is on display and represents both the Student Operators and the University. Leaving the machine in an unclean state because the previous operator did not perform his/her housekeeping duties is not acceptable. Restating, at all times the following rule applies; when the Student Operator leaves the machine, it is to be left in a safe, clean condition.
- Student Operator is responsible for using the machine's safety equipment (provided by the Staff) as well as PPE (provided by the Student Operator's Team) in a manner that is both effective and in accordance with OSHA, PERRP, and industry standard, where applicable. Failure to employ required safety measures because of ignorance of laws, rules and standards is not acceptable and grounds for expulsion from the Facility. When in doubt, ask!
- Upon discovery of any missing guards, damaged equipment or any other form of safety hazards, the Student Operator must immediately cease operation and inform the Shop Supervisor, regardless of fault or involvement.
- Student Operator is responsible for notifying the Shop Supervisor of any crash or accident occurring during operation. Internal damage may not be visible and presents a hidden hazard to other users; the Shop Supervisor must be made aware so that potential problems can be sought out and cumulative wear/damage tracked.
- Student Operator is responsible for properly shutting down and returning the machine to a ready condition for the next operator. Typically this is defined as returning the machine to the state in which you found it, albeit cleaner, with associated tools in their proper locations as specified in SECTION 4k.

## 4. Standard Operation of the Equipment

- **High-Low Range Power Switch**
  The reversing power switch, labelled A in Figure 1 and Figure 7 of Appendix A, controls both power and spindle direction. Thus, it is necessary for the switch selection to correspond with the lever marked O in Figure 2 and Figure 7 of Appendix A for clockwise rotation of the spindle, which accounts for the vast majority of tooling used in the shop. Spinning any tool reverse of its intended direction guarantees it will be destroyed upon contact with a workpiece, potentially creating flying debris and other hazards to the operator. Before switching the machine power on in any direction, ensure that the spindle and tool are free of the workpiece and no indicators or measurement equipment is attached to or in close proximity to the workpiece. Also ensure the cutting tool, if installed, is properly clamped as described in SECTION 4b.

- **Drawbar Operation and Tool Installation**
  For clamping via R-8 collet (depicted in Figure 3 of Appendix A), first ensure that the tool to be clamped is within ±.002" (±.05mm) of the collet's nominal diameter. For determining nominal size, especially on unmarked collets, remember that the vast majority of collets in this Facility are sized on the fractional inch system. After THOROUGHLY cleaning the collet or R-8 compatible adapter, insert the collet or adapter into the spindle nose, labelled K in Figure 7 of Appendix A, and manually spin the drawbar, threading it into the collet. Continue this process until one or two revolutions BEFORE bringing the collet into full contact with the spindle. Insert the tool and hand-tighten the drawbar. Once the drawbar is hand tight, check to see if the tool will stay clamped. If so, proceed by applying a rearward pressure on the braking lever labelled C in Figure 7 of Appendix A and continue to tighten the drawbar using a ¾" wrench. DO NOT OVERTIGHTEN, 25-30 lb-ft of torque on the drawbar should not be exceeded even in the most demanding situations. See the Shop Supervisor with any clamping concerns before proceeding; the spindle and collet surfaces are the heart of the accuracy of the machine and are easily damaged.

- **Speed Adjustment**
  On variable-speed Bridgeport milling heads, the speed of the tool may be adjusted on-the-fly by gently turning the handle marked P in Figure 7 of Appendix A. THIS HANDLE MUST BE TURNED ONLY WHEN THE SPINDLE IS SPINNING. Do NOT adjust the handle while the machine is off, as this will apply tremendous force to the continuously variable belt drive mechanism since it cannot react to the adjustment being made.

  On variable frequency driven machines (speed variation is accomplished via an inverter), a rotary knob is present in the same vicinity as the handle marked P in Figure 7 of Appendix A. This knob will adjust the RPM of the spindle at any time the drive is powered, even while the spindle is not turning. DO NOT adjust the speed to higher than is appropriate for the tooling at any time, even if the machine is off. Were the spindle commanded to spin after an unreasonable RPM was pre-set, this could result in the unexpected and immediate destruction of the tool and the creation of flying debris and other hazards to the operator.

  Do not run any machine for extended periods at high RPM in either high or low range (>2000rpm or >350rpm, respectively) as this causes excessive wear on the components of

the machine. There is generally an easy alternative method to avoid undue stress and wear on the machine accumulated from frequent or extended operation near the machine's limits. See the Shop Supervisor for advice on preferred manufacturing methods.

- **Shifting the Hi-Neutral-Lo Lever**

  This is the most frequently misunderstood system on the machine. The lever is marked O and depicted in Figure 2 and Figure 7 of Appendix A. Please take care to fully understand the order of operations AND the method of confirmation for verifying successful engagement (both Listed immediately below). Restating, the Student Operator MUST learn to verify (with the Shop Supervisor in attendance for the first few times) successful engagement before attempting to shift the Hi-Neutral-Low lever, marked O in Figure 2 of Appendix A.

  SHIFTING TO LOW GEAR – 60-500 RPMS

  - Turn the milling machine power off.
  - You will actuate the shift using the lever that is marked O and depicted in Figure 2 and Figure 7 of Appendix A. FIRST, use your palm to gently press the lever's knob inwards (towards the center of the machine or Y-Z Plane) to disengage the locking pin. SECOND, initially maintain the sideways pressure with your palm and, with your thumb wrapped around the lever, rotate the lever down and back until the lever is more than halfway through the stroke. Release the sideways pressure when you are halfway through the stroke. This is necessary to re-engage the locking pin and allows the pin to 'catch' the low gear position, notifying the operator that the lever has been rotated far enough. DO NOT ROTATE THE LEVER TOO FAR! Rotate it only far enough to engage the locking pin in the LOW position as depicted in Figure 2.
  - Ensure the locking pin is fully seated by gently tugging outward on the lever's knob (away from the center of the machine or Y-Z Plane). DO NOT LET GO UNTIL YOU ARE CERTAIN THE PIN IS FULLY ENGAGED. The lever is heavily spring loaded and can flip forward rapidly, creating a pinching hazard or allowing the gear teeth to clash.
  - With the machine still safely powered off, CHECK ENGAGEMENT by gripping the spindle nose, marked K in Figure 7 and rotating it back and forth. You should feel the backlash in the gears "clicking" quietly. If you do not feel the backlash, the shift was not completed and you must begin the process again. If the spindle spins freely with little resistance, ensure you are not stopping early and putting the machine in neutral. The lever should approximately be at the angle depicted as LOW in Figure 2.
  - The machine, if shifted properly, is now ready to be turned on using the process described in SECTION 4a.

  SHIFTING TO HIGH GEAR – 500-4200 RPMS

  - Turn the milling machine power off.
  - REMEMBER that when the lever is in the LOW position, it is highly spring loaded! Grip the knob fully before disengaging the locking pin so that the lever does not immediately slip from your palm and create a possible pinch hazard or impact the internal gears together! After firmly gripping the lever's knob, press inward and "catch" the lever's movement as the pin disengages.

- SLOWLY allow the lever to rotate forward under the spring torque until the lever stops rotating.
- Because of the mode of engagement and disengagement of the gears, the lever is very likely not to engage when shifting from LOW to HIGH. With your left hand, gently rotate the spindle nose, marked K in Figure 7, back and forth to allow the internal gears to mesh and the shift to be completed. The lever will gently "jump" into position against your hand, notifying you that the gears have begun to mesh and that the lever can GENTLY be used to complete the shift. NEVER use excessive force on the lever to attempt a LOW to HIGH shift completion.
- Ensure the locking pin is fully seated by gently tugging outward on the lever's knob (away from the center of the machine or Y-Z Plane).
- With the machine still safely powered off, CHECK ENGAGEMENT by gripping the spindle nose, marked K in Figure 7 and rotating it back and forth. You should feel the backlash in the gears "clicking" quietly. If you do not feel the backlash, the shift was not completed and you must begin the process again. If the spindle spins freely with little resistance, ensure you are not stopping early and putting the machine in neutral. The lever should approximately be at the angle depicted as HIGH in Figure 2.
- The machine, if shifted properly, is now ready to be turned on using the process described in SECTION 4a.

- **Operating the Quill**

  The Quill, marked J in Figure 7, allows drilling operations to be performed as well as small, course height adjustments for milling operations. The Student Operator must recognize, however, that extending the quill down decreases the rigidity of the machine for operations involving side loads. Milling operations should always be performed with the quill in the most upward position possible to minimize the leverage the tool has over the machine frame, and thereby decrease deflection and potential tool chattering. The Quill Brake, as described below, is to remain engaged at all times, except to move the Quill.

  Operation of the Quill is accomplished by the following;

  - Release the Quill Brake, marked L in Figure 7, by turning the lever counter-clockwise until loose
  - Now that the Quill Brake is released, use the Quill Handle, marked M in Figure 7, to adjust the Quill height to the desired location for milling, or turn the power on using the process described in SECTION 4a and complete the drilling operation.
  - Re-set the Quill Brake by gently turning the lever clockwise until snug. This is a very efficient mechanism, mechanically speaking, and must not be tightened excessively.

  DO NOT PERFORM MILLING (END OR PERIPHERAL) OPERATIONS WITH THE QUILL BRAKE LOOSE. The helix of the mill will act as a screw, drawing the entire quill down and engaging the tool further into the part, leading to a runaway condition that can cause tool breakage and flying debris hazards. Likewise, large drill bits should be used with caution, particularly in pre-drilled holes. The best course of action in this case is to lock the Quill Brake and SLOWLY feed the part towards the drill/mill using the Z axis handle.

- **Table Movement in X,Y,Z**

  The process of positioning the table for measurement, milling or drilling operations is similar for all axes and are listed below.

  The Student Operator must PLAN their movements, both immediate and successive, before laying a hand on the machine! Simply understanding which axis will move when a particular handle is turned is not enough. When you are cutting, you need to follow the cutting plan like a script, where each line of the script includes which handle(s) need to be used, which direction they are to be turned to produce and desired movement and which brakes need to be released prior to movement and re-set after the desired position is reached. Accidently grabbing the wrong handle or momentarily turning the right handle the wrong way can lead to immediate disaster for the cutter and the part that the Student Operator intends to make. This can lead to flying debris hazards as well as damage to the machine, not to mention the part and cutter.

  Making a dry run (mentally or even physically, i.e. while the cutter is at a clearance height above the workpiece) through all operations up to the next pause point is critical and will reduce the human error factor, drastically improving not only the safety, but also the productivity of the Student Operator.

  X-Axis Movement;

  - The X-Axis brake(s) are located on the front face of the saddle, marked “Table Clamp Levers” in Figure 6. Depending on the machine, there may be one or two brakes for the X-Axis. Turn the Brake Lever counter-clockwise until the torque to turn the handle drops significantly. This disengages the brake, DO NOT continue to turn the handle counter-clockwise after the brake is disengaged.
  - Adjust the X-Axis to the desired position by turning the handles located on the left and right ends of the table. Clockwise rotation of the handle moves the table towards the user. Counter-clockwise rotation moves the table away from the user.
  - As you are nearing the desired X axis position, begin to re-apply the X-Axis brake by GENTLY turning the X-Axis Brake Lever Clockwise until snug. DO NOT OVERTIGHTEN, the lever should be able to readily disengage with one hand. You should plan to reach the desired X-Axis position and full brake position at the same time. This prevents the table from shifting position while the brake is applied, which, if occurring after the desired X-Axis position is reached, represents error that if not noticed by the Student Operator can negatively affect the workpiece or induce a tool breakage if working close to the fixture or workpiece surfaces not intended to be cut.

  Y-Axis Movement;

  - The brake for the Y-Axis is located on the left side of the saddle and is to be operated the same as listed above for X-Axis Movement.
  - Move the Y-Axis as described above for X-Axis Movement using the front-most handle on the machine. As with the X-Axis, clockwise rotation of the handle moves

the table towards the user. Counter-clockwise rotation moves the table away from the user.

- Reset the Y-Axis brake as described above for Y-Axis Movement.

Z-Axis Movement;

- The brake for the Z-Axis is not always a machined knob as used on the X-Axis and Y-Axis. On Bridgeport and Bridgeport Clone Milling Machines, the Z-Axis brake lever is a simple rod threaded into the brake body. On these mills, it is located under the Z-Axis handle. On other styles of manual milling machines the Z-Axis brake(s) are located on the left side of the knee, near the dovetail way. Generally, there is enough friction in the way and lifting screw that the table does not move even without brake(s) applied, but for precision applications or heavy milling, the brake(s) should be applied as described above for X-Axis Movement.
- After ensuring the Z-axis brake(s) are DISENGAGED, the operator is free to adjust the Z-axis position by cranking the handle that protrudes from the knee at an angle from the left side of the knee. The Z-Axis handle is indexable by sliding the handle towards the operator, thus disengaging the driving splines visible near the dial. The Student Operator MUST ensure that the handle stays engaged during adjustment of the Z-Axis position. This can be ensured by stopping every few rotations and gently pushing the handle towards the machine before resuming Z-axis adjustment.

- **Drilling**

Drilling operations, when performed properly, generate only axial vertical forces and torque from cutting. The torque due to typical drilling (Ø<.500”) is generally quite low, thus, the clamping arrangement is more flexible at smaller diameters.

Drill Chucks, as depicted in Figure 4, can be used for drilling hole diameters below $\frac{1}{2}" / 13mm$. Drill bits with reduced diameter shanks having a drilling diameter over $\frac{1}{2}" / 13mm$ MUST be clamped directly in a collet (as depicted in figure 3) to prevent slippage which would damage a drill chuck and render it inaccurate and unsafe for future use. See ‘Section 4 - Drawbar Operation and Tool Installation’ for instructions for both clamping directly using a collet as well as clamping a drill chuck mandrel in a collet.

REMEMBER – Collets have >90% contact to the tool shank and can more rigidly clamp tooling. Drill chucks, on the other hand, only engage with a line of contact under each of the three jaws. This means that drill chucks induce much higher stresses on the tool shank and are much more prone to slippage or misalignment. Thus, though drill chucks are convenient due to their infinite adjustability, they must be used only when the loads are appropriately low. Conversely, the selection of R8 collets is limited in the shop and special size collets must be purchased by the Student Operator’s team. These R8 collets can be found on the McMaster Carr website and, if ordered by 6pm, will arrive the next day. Not having the proper collet is NOT an excuse to abuse the nearest size collet or to clamp a tool that will generate high loads in a drill chuck. In either of these cases, tool slippage/breakage is highly likely and can generate flying debris hazards as well as damage to the workpiece and the machine.

Student Operators needing to cut large diameter holes in a workpiece are often tempted to use a hole-saw clamped in the spindle of a milling machine. There *are* means to properly do this, however Student Operators are PROHIBITED to attempt this operation without the presence and express permission of the Shop Supervisor. See the Shop Supervisor well in advance so that proper tooling can be acquired and the process planned.

- **End Milling, Peripheral Milling, Slotting**

  To classify several types of typical milling processes;

  - End-milling generally includes shallow cutting operations (Depth of Cut <10% x Mill Diameter) that engage nearly the full cutting width of the mill. This includes facing operations
  - Peripheral Milling involves cutting operations were the forces are unbalanced due to material being cut only with one side of the endmill. This includes trim-to-length operations
  - Slotting operations engage the full cutting diameter of the endmill. During slotting, depth of cut may be greater than with end-milling and sidewalls are near or in contact with the endmill.

  Guidelines applicable to all milling processes;

  - Forces generated include radial and axial. Radial forces generated by the leading edge of the cutting action may cause deflection of the endmill, “tipping” the endmill so that one side cuts lower than the opposite. This generally is not a hazard to the user, but may create a surface that has a “sawtooth” or triangular pattern instead of being flat. The operator should reduce depth of cut and/or feedrate to mitigate this effect.
  - Tool MUST be clamped directly in a collet. NEVER clamp an endmill in a drill chuck
  - The Quill Brake must be engaged – See Section ‘Section 4 – Operating the Quill’. If the quill brake is not engaged, the helix of the endmill flutes will cause the cutting action of the endmill to “pull” the cutter/collet/spindle/quill down into the part, resulting in a heavier cut that pulls the aforementioned assembly down even harder. This can result in a runaway condition that will result in tool breakage/workpiece destruction as well as creating the potential for flying debris hazard to the operator.
  - Axes not actively being adjusted must have their respective brakes engaged - See Section ‘Section 4 – Table Movement in X,Y,Z’
  - The Student Operator must be able to recognize the maximum safe RPM and feedrate. RPM limits are set based on workpiece material or cutting tool material heat limits as well as the cutting operation. Feedrate is determined by the cutting tool geometry (Number and rake of cutting flutes) as well as the diameter (bending strength) of the endmill. The feedrate may also be limited by the geometry/rigidity of the part in situations involving parts with unsupported/thin geometry or brittle materials.

  Guidelines applicable to peripheral milling processes;

- Forces generated include radial and axial. Radial forces are generally stronger due to the increased side contact on the endmill and may deflect the endmill away from the surface, leaving a pronounced taper to the surface. This is an indication that too heavy of a radial cut is being taken. Reduce the radial depth of cut and/or feedrate to mitigate this effect.
- The operator should also remember that increasing the axial depth of cut can create very large axial pull forces on the cutter/collet/spindle/quill assembly. High axial depths of cut should be combined only with very light radial depths of cut to minimize this pulling action. If the student suspects the endmill is being pulled down from the collet, or the brake of the quill is not able to prevent the downward movement of the quill, check that the drawbar and quill brake have been set properly as described in Section 4 - Drawbar Operation and Tool Installation and Section 4 - Operating the Quill. DO NOT overtighten either the drawbar or quill brake to compensate. Reduce axial and/or radial depth of cut and/or reduce feedrate to lessen the pulling action.
- Climb and Conventional milling produce drastically different results during peripheral milling. Climb milling is to be avoided for heavy axial or radial depths of cut. Very light passes (radial depth of cut <.005", axial depth of cut < mill diameter) must be used in conjunction with light table axis brake pressure, as described in Section 4 – Table Movement in X,Y,Z to prevent tool destruction, workpiece and milling machine damage, and the resulting flying debris hazard. Heavier passes than those described above MUST use conventional milling.

Guidelines applicable to slotting processes;

- Slotting is essentially end-milling while surrounded by the recently cut walls of the slot in the workpiece. This requires the Student Operator to take special care to evacuate the swarf/chips made during the cut. Letting the cutting tool re-cut these chips is not allowed. Air or periodically retracting the cutting tool and powering down the machine for manual brush removal of the chips is necessary to prevent the old chips from being re-cut or from clogging the exit path of new chips and potentially causing a tool breakage.
- The axes not involved in the slotting operation MUST have their table axis brakes properly engaged to ensure the cutter does not drift into the walls of the slot and cause a sudden and unexpected drastic increase in the cutting forces. See Section 4 – Table Movement in X,Y,Z for more details on table axis brakes.

- **Flycutting**

  Flycutting is a facing operation meant to provide an inherently flat surface good for mating other machined parts to or for making sealing surfaces. It involves the use of a single flute (indexable carbide or HSS) cutter mounted in a special mandrel that is clamped via collet in the machine spindle. This type of cutter must be used with caution for the following reasons;

  - Extended length tools are subject to bending, vibration and can consequently easily reach a resonance or "chatter"

- Extended length tools present a wrap/pull-in point, cutting point and thrown object hazard to the operator. The Student Operator MUST ensure chips, tools, loose parts, machine guards and body parts will remain clear of the cutter during operation. This check must be performed prior to powering up the machine. This can be safely physically verified by ensuring the machine is powered off, placing the machine in neutral and manually turning the cutter to check clearance to the fixture, guards, etc.
- Due to the extended length/diameter, a flycutter experiences vastly higher cutting speeds for a given RPM than typical end mills. The Student Operator MUST verify the proper cutting speed prior to attempting the operation.

- **Required Maintenance – Pre and Post Inspection**

  BEFORE operating the machine, the Student Operator is responsible for the following;
  - Ensure that the floor mats are swept clean and lay flat on the floor without creating a trip hazard or impeding access to safety features of the machine.
  - Be sure to remove tools, parts, swarf/chips, etc. that previous operators may have left on the machine and that operation of any feature of the machine will not be impeded by objects sitting nearby.
  - The axis guides/areas of sliding contact of the machine must be gently BRUSHED clean. Do this in a way that moves chips away from sealing surfaces and DO NOT use compressed air to clean the exposed axis guides/dovetails/linear slides as this tends to wedge chips/dust/swarf deeper into the sealed regions of the machine.
  - If the machine is equipped with automatic oiling system(s), check that the fluid levels in the lubrication reservoirs are acceptable.
  - If the machine is equipped with manual or spring-return oiling systems, make sure to actuate those prior to moving the machine to prevent metal-on-metal contact in the sliding portions (“ways”) of the machine.

  AFTER operating the machine, the Student Operator is responsible for the following;

  - Ensure the tools necessary for the machine are back in their proper locations.
  - Clean the machine and area of ALL swarf/chips.
  - Make sure to clean under the floor mats as well as sweep the swarf/chips from the surface and that the floor mats lay flat without creating a tripping hazard.
  - Ensure that machine guards are in place and in good condition for the next operator.

## 5. Personal Protective Equipment

### a. Required PPE

- Clear eye protection, in good condition, must be worn at all times in the Facility to guard against flying debris. In situations where shaded eye protection (goggle or glasses) is necessary to avoid exposure to radiation sources, clear eye protection may be supplemented with the appropriate shaded eye protection, however, clear eye protection must still be worn under welding helmets.
- Face shields, fire-retardant aprons and gloves may be necessary in certain operations producing hot or flying swarf/chips. However, wearing gloves may create additional hazards around pull-in and pinch points. See the Shop Supervisor for advice on appropriate PPE in special cases.
- Closed toed shoes and long pants must be worn at all times in the Facility.
- Clothes having a high cotton content are highly recommended. Many athletic/sport oriented synthetic clothing has a low melting temperature and can cause serious burns to the operator's skin if exposed to heat sources like hot workpieces, cutting tools and swarf/chips.

### b. PPE Storage

- The teams in the Facility are individually responsible for ensuring the availability of PPE for the Student Operators.
- Specialty PPE for one-time operations may be requested/borrowed. See the Shop Supervisor in these cases.

### c. PPE Inspection

- The teams are required to ensure that, at all times, appropriate PPE is available for the operations the team regularly performs.
- The teams must keep enough quantity of the appropriate PPE to ensure that all members participating in a given operation have PPE available to them.

## 6. Required Training

All operators of the manual milling machines located at the Facility are REQUIRED to have documented training. Documentation must include facility activity disclosure forms as well as machine training. The training must be conducted by the Shop Supervisor and be on file at the Facility BEFORE the operator is permitted access to the machine. The training will cover the operation of the manual milling machine as well as, including but not limited to, the following topics;

- High-Low Range Power Switch
- Drawbar Operation and Tool Installation
- Speed Adjustment
- Shifting the Hi-Neutral-Lo Lever
- Operating the Quill
- Table Movement in X,Y,Z
- Drilling
- End Milling, Peripheral Milling, Slotting
- Flycutting
- Required Maintenance
- Understanding of the SOP that is written
- Understanding of the Job Hazard Analysis / Required PPE
- Common causes or misuse that can lead to damage or injury
- Pre-Inspection and what to do if there is a safety issue or service issue needed for the equipment
- Emergency Procedures

## 7. Emergency Procedures

**a. Equipment Malfunction/Tool Breakage/Fixturing Failure**

Upon any failure of the equipment, tool, or fixturing, immediately remove the tool from contact with the part and turn the machine off using the method described in SECTION 4 - High-Low Range Power Switch. Removal of tool contact is usually possible by retracting the Quill, as described in SECTION 4 – Operating the Quill. Alternatively, the Table can be lowered quickly by adjusting the Z axis handle as described in SECTION 4 – Movement in X,Y,Z. Before turning the machine on again, check to ensure the following;

- Check that the Tool is clamped securely and is not damaged from the incident
- Check that the Part is clamped securely: If the part becoming loose was the reason for the stoppage, DO NOT PROCEED by re-fixturing in the same manner. Contact the Shop Supervisor for advice or seek assistance from more experienced Student Operators.
- Check that the machine is safe to turn back on: If the machine did not perform as expected, i.e. spindle rotation reversed suddenly, spindle rotation stopped or any handle required higher than usual force input, contact the Shop Supervisor before proceding.

**b. Building Evacuation/Announced Emergency**

Cease operations immediately by removing tool from contact and turning off the power to the machine as described in SECTION 4 – High-Low Range Power Switch and SECTION 4 – Operating the Quill/4 – Moving the Table in X,Y,Z. Ensure that any items capable of generating immediate danger, i.e. flammables, sharp protruding objects, etc., are taken care of as best as possible in an expedient manner and calmly exit the building or follow the directions of the most senior person in the Facility.

**c. Power Outage/Unexpected Emergency**

During a power outage or heavy storms, all equipment should be turned off to prevent mechanical and electrical hazards to the Student Operators in the building. The equipment should be confirmed to be switched off so that, in the event of a power outage and subsequent power restoration, the equipment does not suddenly and unexpectedly start up. During heavy storms, the operations should be ceased and the equipment should be powered off for the duration of the storm so that the operator doesn't experience an unexpected stoppage that could result in tool breakage and personal injury.

## 8. Lock Out/Tag Out

This manual vertical milling machine is equipped with a lockout compatible power disconnect switch mounted on the wall behind the machine. If color-coded lock-out equipment is present on the disconnect switch, preventing the powering of the machine, the machine is to be considered removed for operation until the person who installed the lock-out gear removes it after confirmation of completed repairs to the machine. DO NOT attempt to remove the lock-out equipment or alternatively power the machine until the machine is reintroduced to service by the lock-out tag-out authorized personnel.

As of the following date, the people listed below are the sole authorized personnel allowed to perform lock-out tag-out operations. This includes the act of locking out to remove from service as well as the act of lock removal and reintroduction to service.

As of 1-1-17, the following are authorized lock-out tag-out personnel;

- Darrin Orr
- Matthew Little
- Don Williams

Any Student Operator who has reason to believe a machine needs service for any reason, safety or otherwise, is responsible for contacting the Shop Supervisor immediately.

## Appendix A

Figure 1

Figure 2

Figure 3

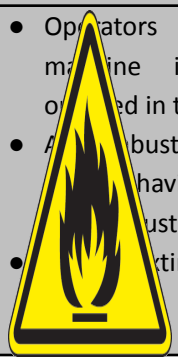
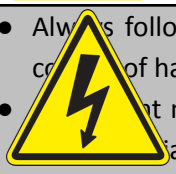
Figure 4

Figure 5

Figure 6

Figure 7

## Appendix B - Job Hazard Analysis

| Work Duties | Potential Hazards | Controls |
|---|---|---|
| Operating the Machine | **Flying Debris** |  • Saf[illegible] glasses are required while o<br>• M[illegible]e mounted guards must b[illegible] [illegible]riate manner |
| Operating the Machine | **Pinch, Crush and Entanglement Hazards** |  • Ref[illegible] to SOP and training fo[illegible] pr[illegible]dures.<br>• U[illegible]aution while handling heavy c[illegible]s |
| Operating the Machine | **Entanglement, Lacerations and Flying Debris Hazards** | • During Tool Changes, machine mu<br>• Operator must be trained in th[illegible] use of the brake to prevent [illegible] startup<br>• Never reach towards the [illegible] operation while machine is runn |
| Operating the Machine | **Fire** |  • Op[illegible]ators are trained to shut d[illegible] ma[illegible]ine in accordance with [illegible] o[illegible]d in the machine-specific SOP<br>• A[illegible]bustibles must be clear of th[illegible] [illegible]havings, combustible dust [illegible]ust be cleaned up appropria[illegible]<br>• [illegible]xtinguishers must be acces[illegible] |
| Operating the Machine | **Slips, Trips and Falls** |  • Proper housekeeping must be m[illegible] at all times, including cleaning [illegible] immediately |
| Machine Service | **Electrocution/Accidental Start up** |  • Alw[illegible]s follow Lock out / Tag out c[illegible] of hazardous energy<br>• [illegible]t needing serviced shall [illegible]ately by authorized per[illegible]<br>• Service should be completed in [illegible] the manufacturer requirements b[illegible] and authorized personnel or a cont[illegible] |

| | | |
|---|---|---|
| Post-Operation Housekeeping | **Lacerations or Foreign Body in Eyes** |  <br>● W[illegible]n cleaning up safety glasses a[illegible]e worn at all times<br>● [illegible]ttempt cleaning up swarf/c[illegible]s running<br>● Gloves should be used with cautio machinery, as they may pre entanglement hazard |